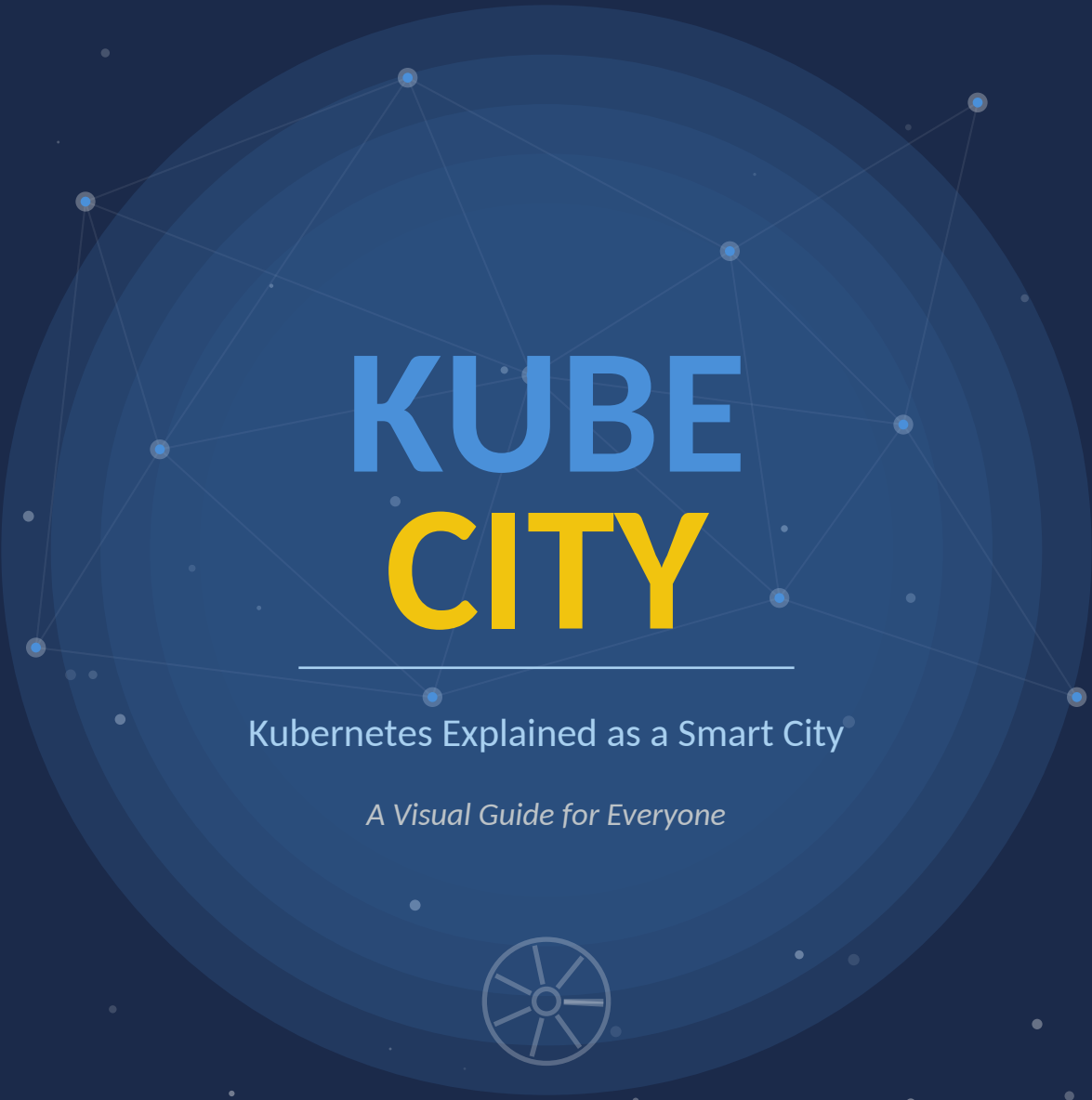




KUBE CITY

Kubernetes Explained as a Smart City

A Visual Guide for Everyone

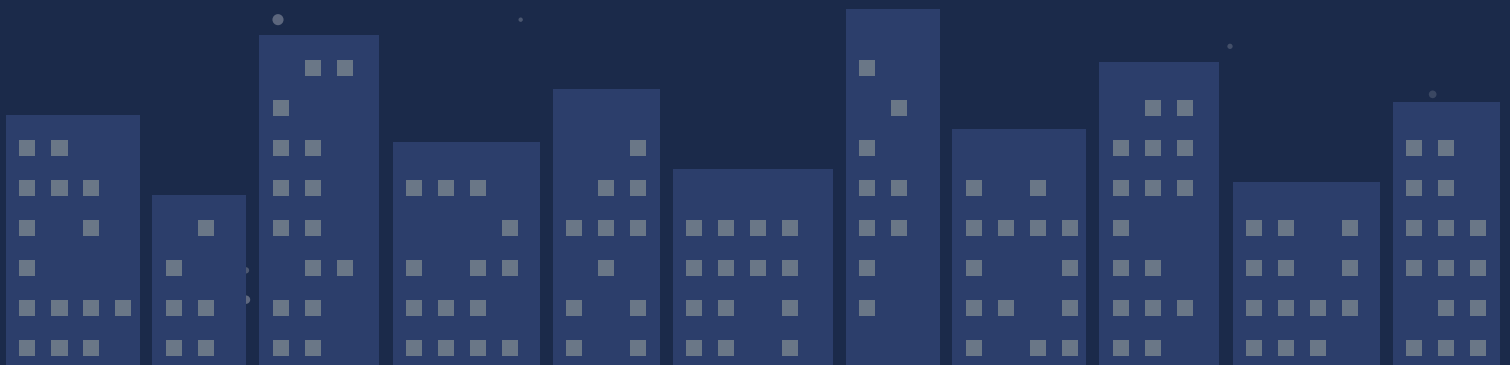


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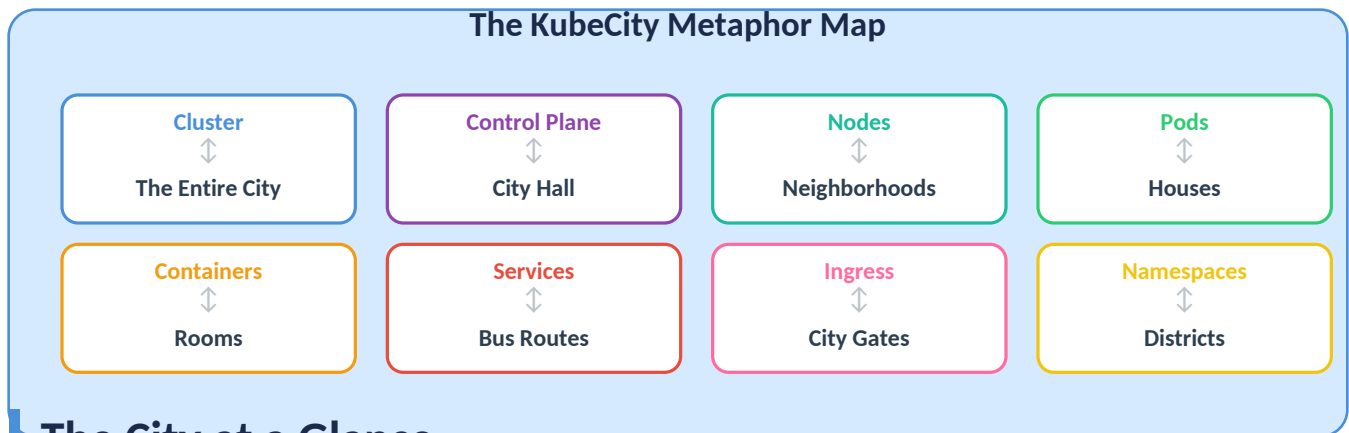
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Complete Architecture

What is Kubernetes?

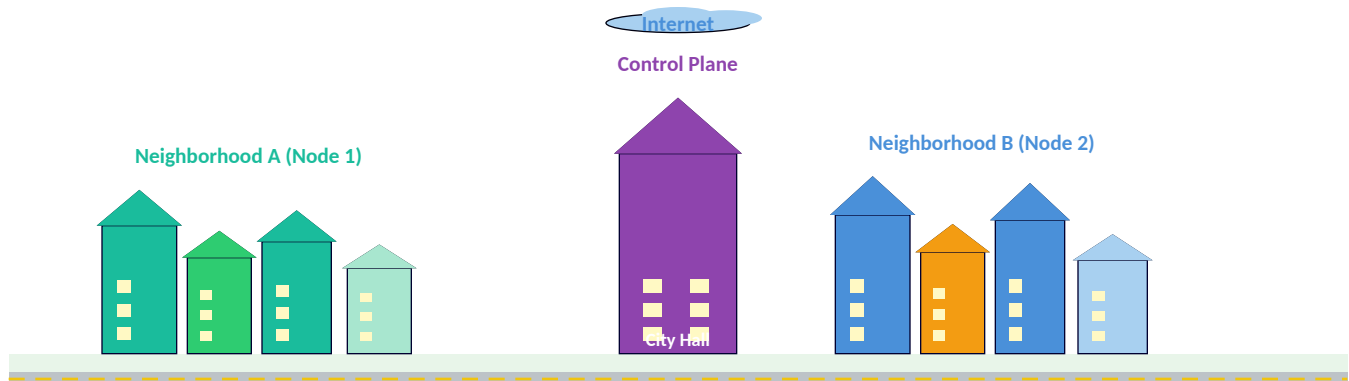
The Short Answer

Imagine you run an ENTIRE CITY. You need to make sure every building has power, every road works, every citizen is safe, and if a building falls down, a new one gets built automatically. That is what Kubernetes does — but for computer programs.

Kubernetes (often shortened to K8s) is like a SUPER-SMART CITY MANAGER. It takes your apps (the citizens), puts them in containers (houses), spreads them across servers (neighborhoods), and makes sure everything keeps running — even when things go wrong.



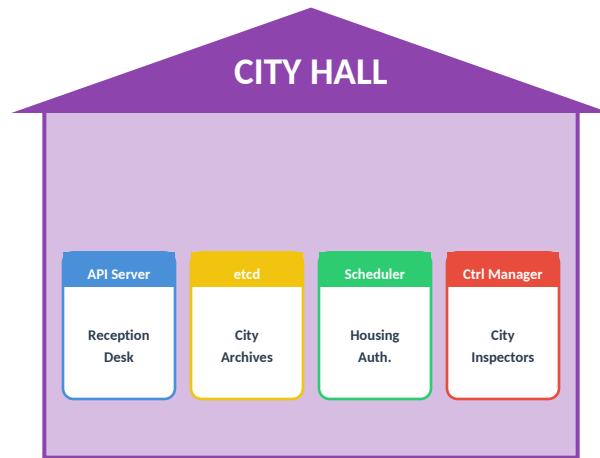
The City at a Glance



The Brain of the City

Where all decisions are made

Every city needs a City Hall where the mayor and city officials work. In KubeCity, the Control Plane is our City Hall. It does not run any citizen apps — its only job is to **MANAGE** the city. If City Hall goes down, the city keeps running, but no new decisions can be made.



API Server = Reception Desk

The FRONT DOOR. Every request goes through here.

etcd = City Archives

The RECORD ROOM storing ALL city information.

Scheduler = Housing Authority

Decides WHICH neighborhood a new citizen moves into.

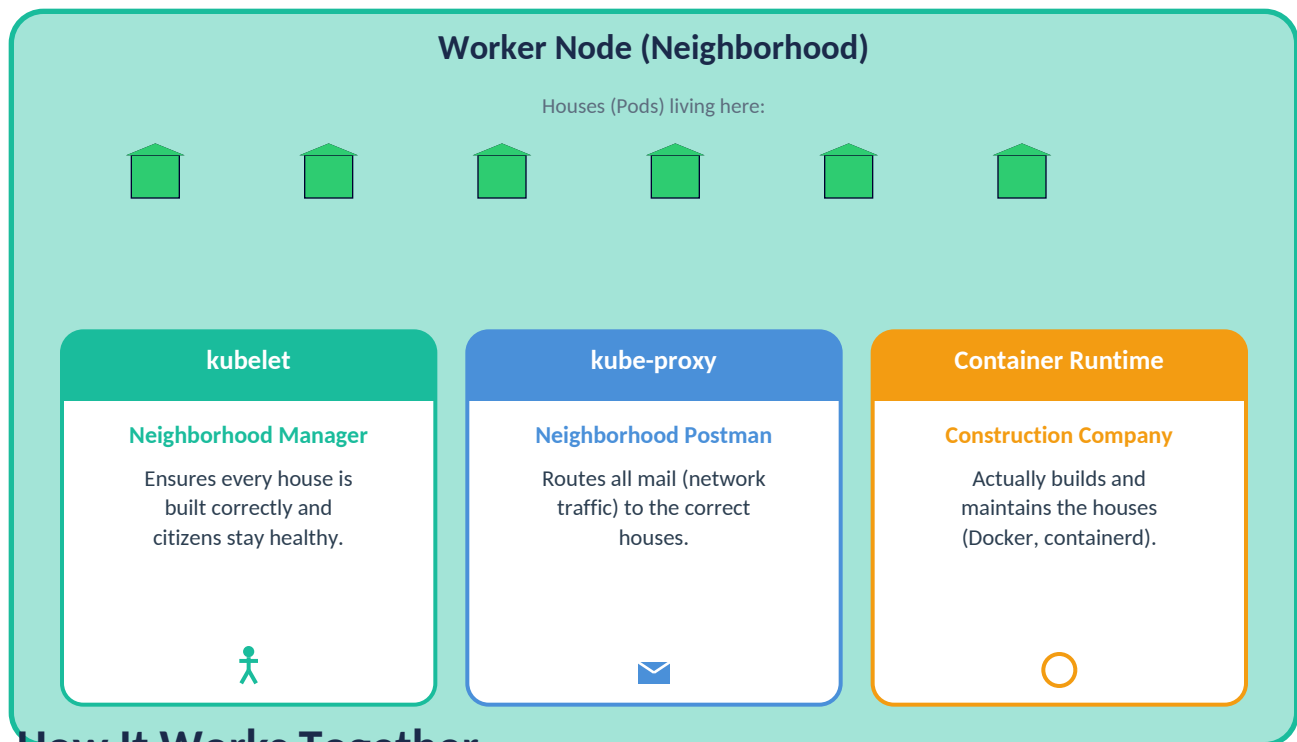
Controller Manager = Inspectors

Walk around making sure everything matches the plan.

Where Citizens Actually Live

The Worker Nodes

While City Hall makes decisions, the actual living happens in Neighborhoods (Worker Nodes). Each neighborhood has its own local manager, postman, and construction company.



How It Works Together

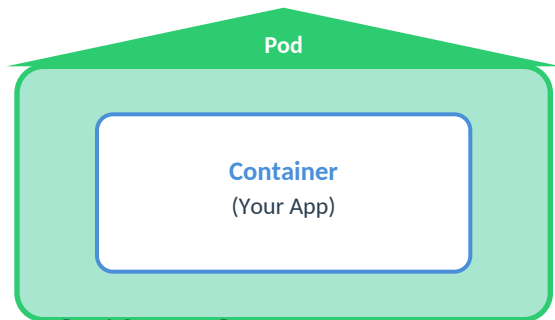
When City Hall decides a new citizen needs a home, the Scheduler picks the best neighborhood. The kubelet tells the Container Runtime to build a house. The kube-proxy makes sure mail reaches it.

The Smallest Living Spaces

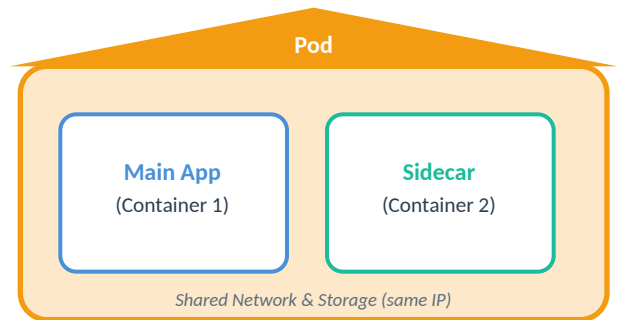
Pods & Containers

A Pod is like a HOUSE — the smallest unit where your app can live. Inside each house, there can be one or more ROOMS (Containers). Most houses have one room, but some have multiple rooms sharing network and storage.

Single-Room House (1 Container)



Multi-Room House (Sidecar)



Pod Lifecycle

The life of a house in KubeCity



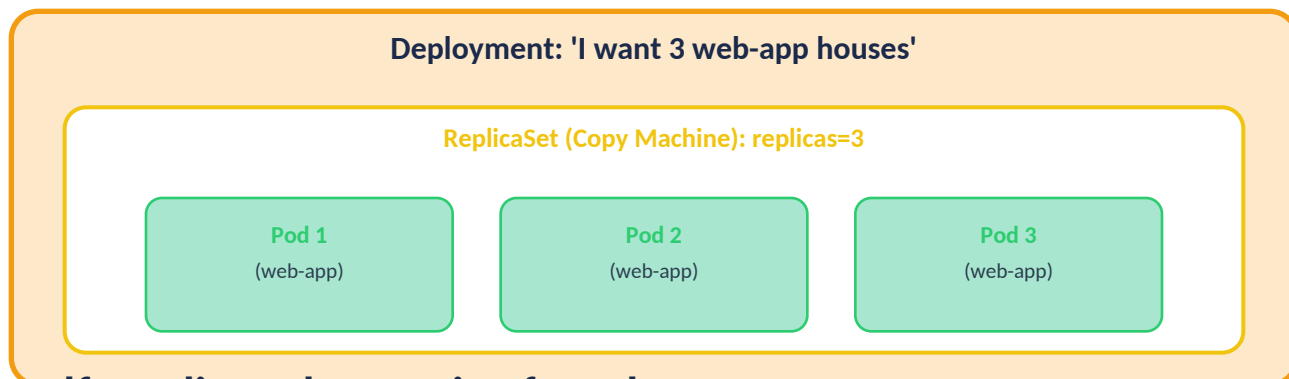
Key Insight:

Pods are TEMPORARY — like pop-up houses. When a pod dies, K8s does NOT repair it. It builds a brand new house. This is why we need Deployments to keep rebuilding them!

Blueprints That Build Themselves

Deployments & ReplicaSets

A Deployment is like a BUILDING PLAN: 'I want 3 identical houses at all times.' If one falls down, it automatically builds a replacement. A ReplicaSet is the copy machine.



Self-Healing: The Magic of Deployments



Rolling Updates: Zero Downtime

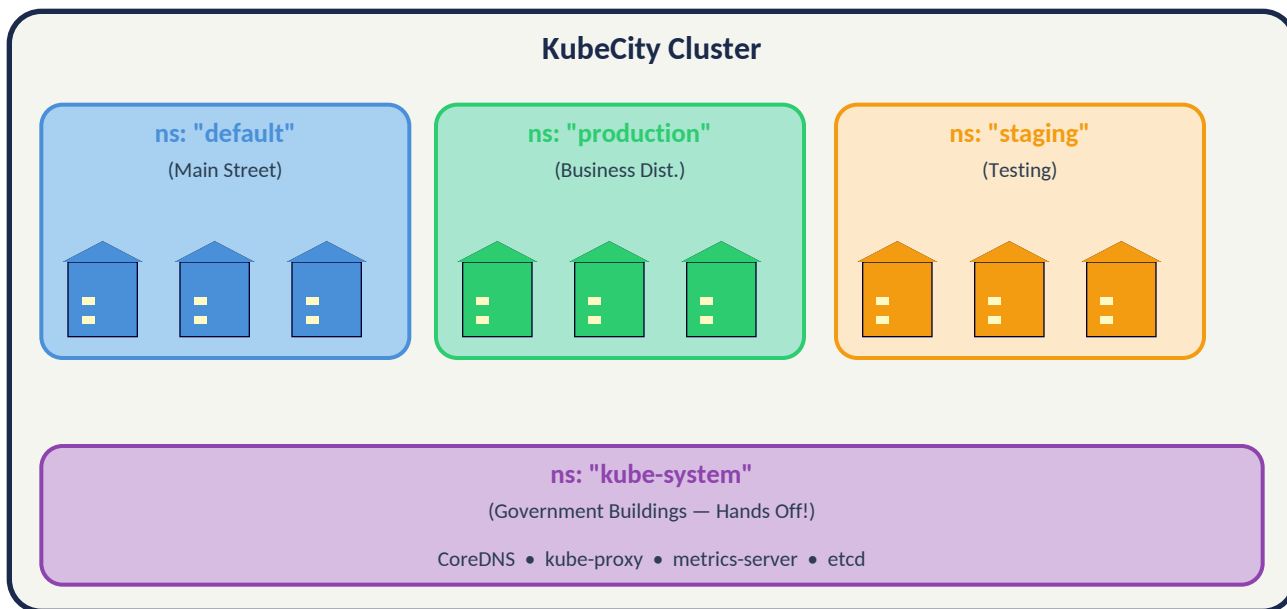
Kubernetes builds new houses one at a time and moves citizens over — zero downtime!



Organizing the City Into Zones

Namespaces

A big city has districts — Downtown, Business, Industrial. Namespaces divide your cluster into separate zones so teams and apps don't collide.



Namespace Rules of the City

Same Name, Different District:

Two apps can be called 'web-app' in different namespaces.

Budget Limits:

Each district can have a ResourceQuota to prevent resource hogging.

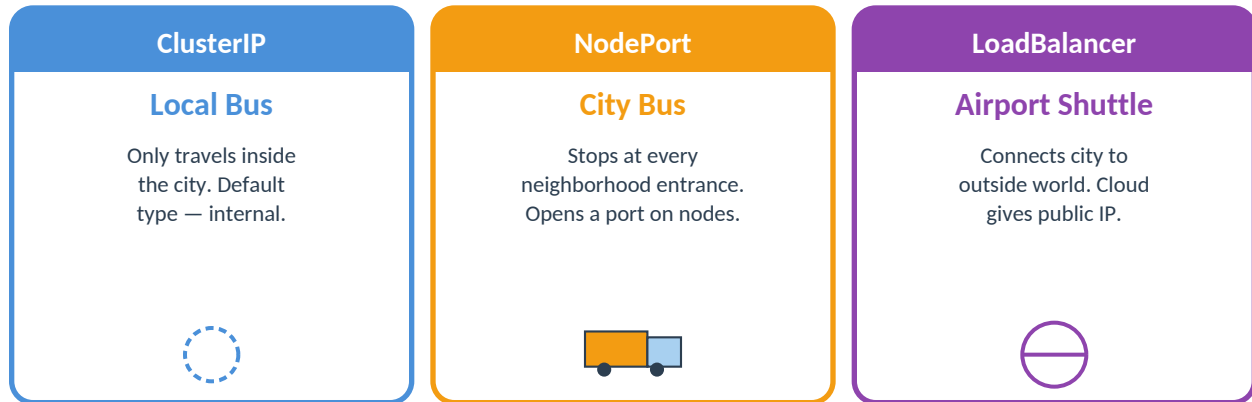
Security Fences:

Add NetworkPolicies to restrict traffic between districts.

How Citizens Get Around

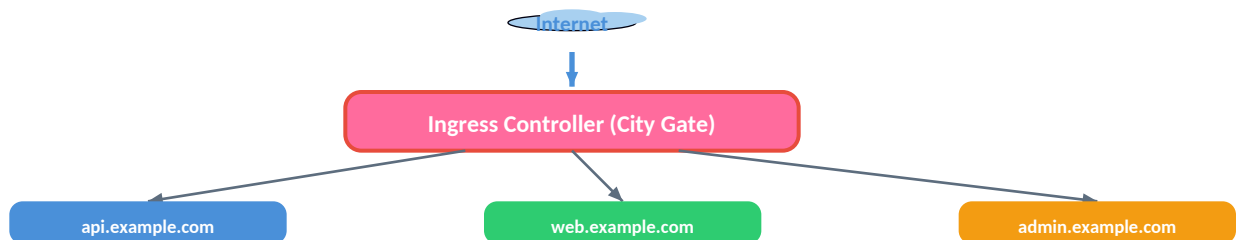
Services & Ingress

Pods get new addresses when rebuilt! A Service is a permanent BUS ROUTE with a fixed name. Take the 'web-app' bus and it always delivers you to the right place.



The City Gate: Ingress

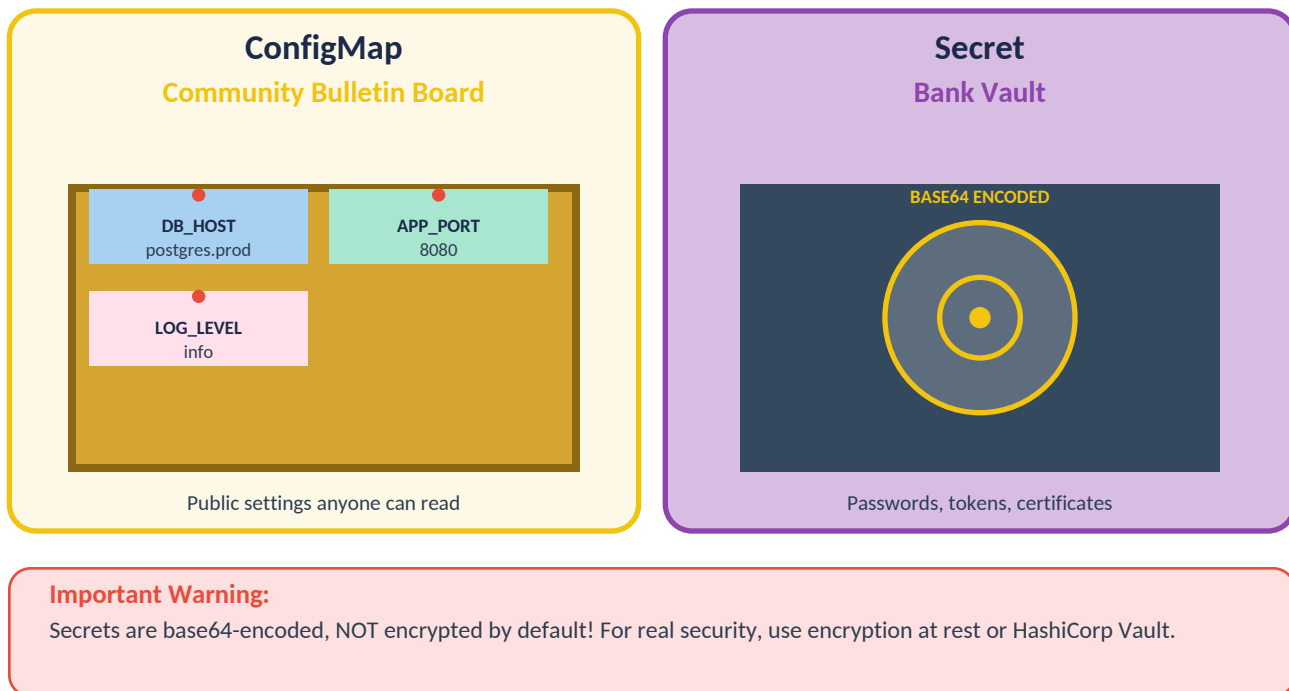
An Ingress is the MAIN GATE with smart signboards. When a visitor asks for 'shop.example.com,' the gate routes them to the right Service.



City Information Systems

ConfigMaps & Secrets

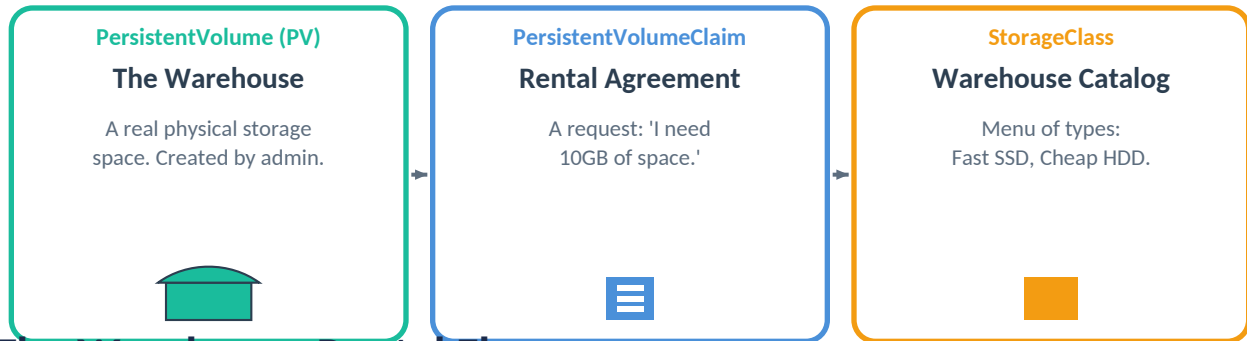
Apps need settings and passwords. KubeCity has Bulletin Boards (ConfigMaps) for public settings and Bank Vaults (Secrets) for sensitive data like passwords.



Where Valuables Are Kept Safe

PV, PVC & StorageClass

Pods are temporary — when destroyed, contents are lost! Persistent Volumes are STORAGE WAREHOUSES outside the house. Your data survives even when houses are torn down.



The Warehouse Rental Flow

- 1 Admin creates warehouse (PV) or enables auto-provisioning
- 2 App requests space via rental agreement (PVC)
- 3 Kubernetes matches request to available warehouse
- 4 Pod mounts warehouse — data survives pod death!

Warehouse Access Modes:

ReadWriteOnce (RWO)	Private locker — one tenant
ReadOnlyMany (ROX)	Museum — many viewers
ReadWriteMany (RWX)	Shared workspace — everyone

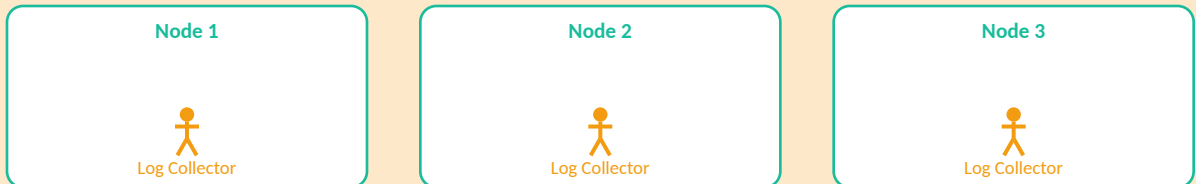
Essential City Operations

DaemonSets, Jobs & CronJobs

Every city needs utilities — trash collection everywhere, one-time deliveries, and scheduled services. Kubernetes has special workload types for these.

DaemonSet = Utility Services

One worker in EVERY neighborhood



Job = Delivery Truck

One-time task, then done



DB Migration • Batch Export

CronJob = Scheduled Service

Repeating on a schedule



Nightly Backups • Log Rotation

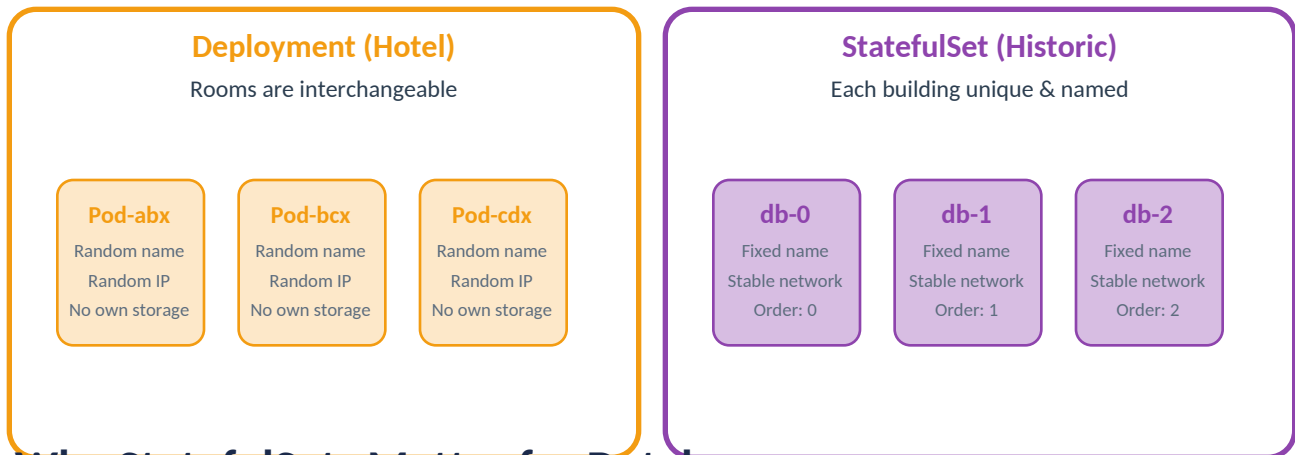
Quick Comparison

Feature	DaemonSet	Job	CronJob
Analogy	City Utility	Delivery Truck	Garbage Truck
Runs On	Every node	Any node	Any node
Frequency	Always	Once	On schedule
Example	Monitoring	DB migration	Nightly backup

When Identity Matters

StatefulSets

Regular pods (from Deployments) are interchangeable — like hotel rooms. But some need a PERMANENT ADDRESS — like a hospital or bank. StatefulSets create numbered, named buildings that keep their identity.



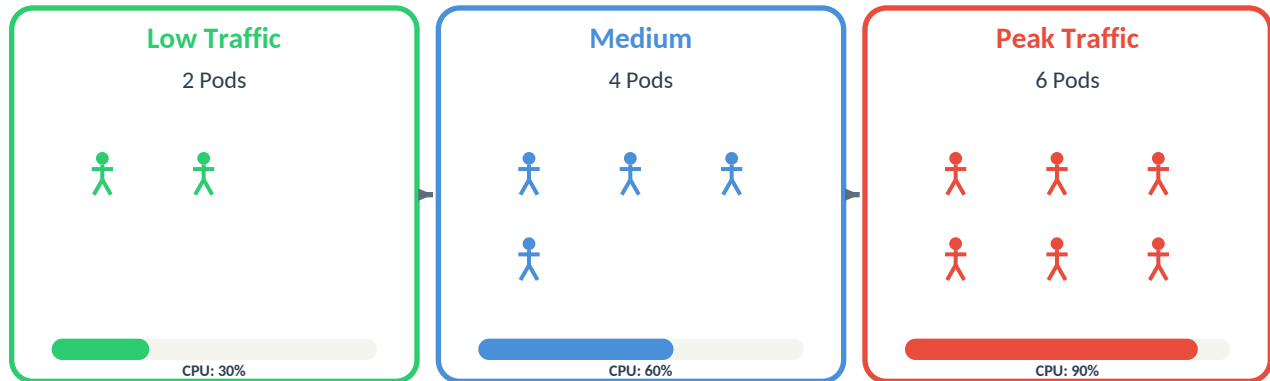
Why StatefulSets Matter for Databases

- **Stable Identity:**
PostgreSQL primary must always be 'db-0'. Replicas know it by name.
- **Ordered Startup:**
Primary (db-0) starts FIRST, then replicas join in order.
- **Persistent Storage:**
Each database keeps its own dedicated PVC that survives restarts.
- **Stable DNS:**
Predictable: db-0.db-svc.namespace.svc.cluster.local

Growing and Shrinking With Demand

Horizontal Pod Autoscaler

KubeCity's coffee shop has 2 baristas on Monday. On Saturday a HUGE crowd arrives! The HPA sees the line and AUTOMATICALLY hires more. When the crowd leaves, extras go home.



Types of Scaling

Horizontal Pod Autoscaler (HPA)

Add more baristas when busy, remove when quiet.

Vertical Pod Autoscaler (VPA)

Give each barista a bigger machine (more CPU/mem).

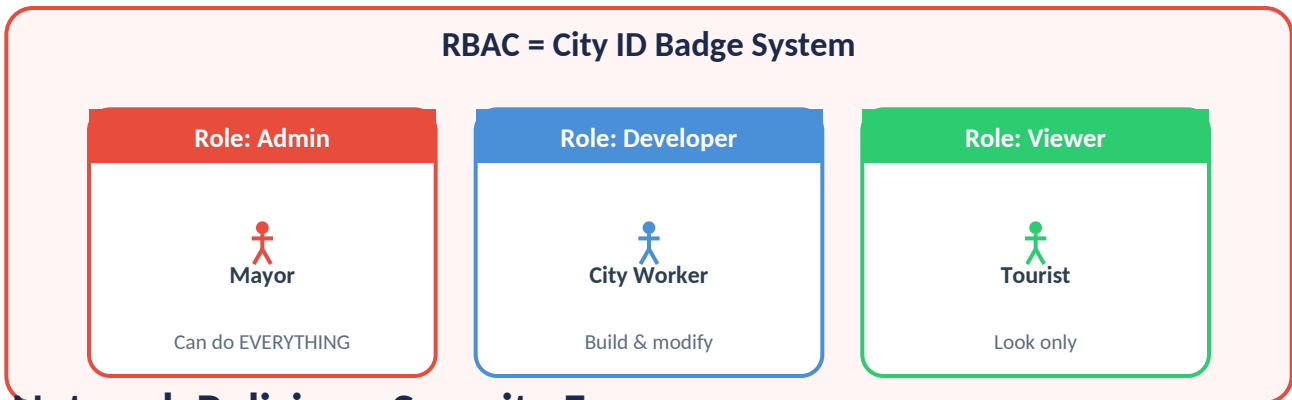
Cluster Autoscaler

Build new neighborhoods (nodes) when all are full.

Keeping the City Safe

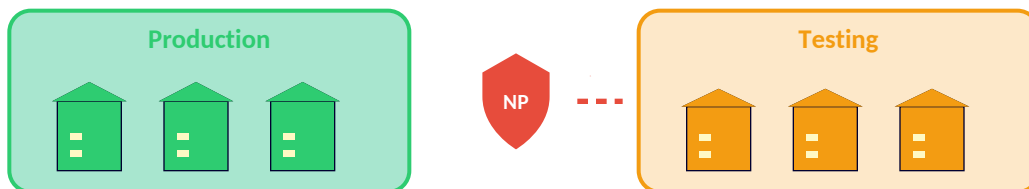
RBAC & NetworkPolicies

Not everyone should have keys to every building! RBAC is the city's ID BADGE system. Network Policies are SECURITY FENCES between districts.



Network Policies = Security Fences

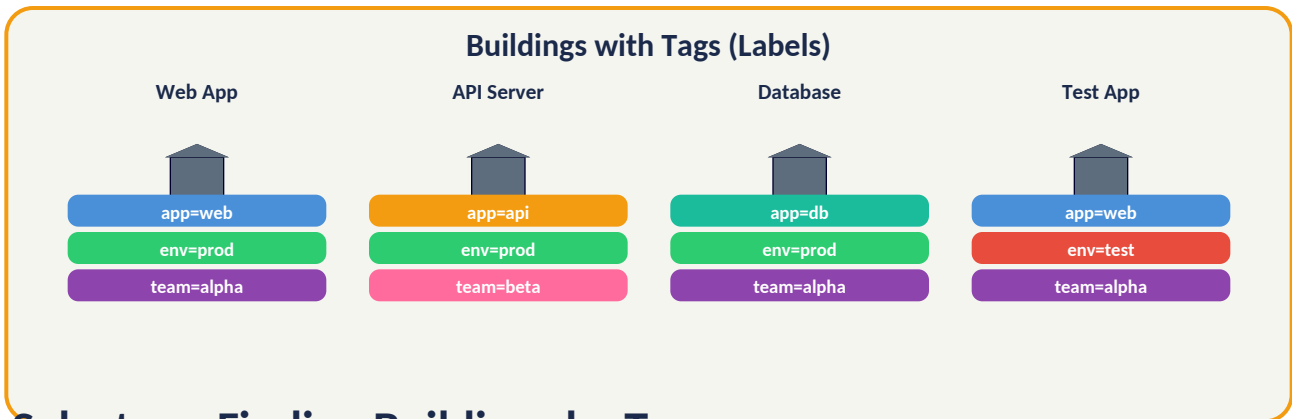
By default, all neighborhoods can talk freely. Network Policies add checkpoints to block unwanted traffic between districts.



Color-Coded City Organization

Labels & Selectors

Every building has color-coded TAGS — 'restaurant,' 'Team Alpha,' 'version 2.' Labels are key-value tags on any K8s object. Selectors FIND things by tag.



Selectors: Finding Buildings by Tag

When a Service says 'app=web, env=prod,' it uses a Selector. Only matching buildings respond!

Selector: app=web, env=prod

Matches 'Web App' only — 'Test App' has env=test, no match.

Annotations: Sticky Notes

Annotations are detailed STICKY NOTES — metadata not used for finding things, but useful for tools. Example: 'Last updated by John on March 5.'

Advanced City Planning

Helm Charts & Operators

Helm = City Blueprint Store

Pre-made plans. Want PostgreSQL? Pick the chart. Need monitoring? Grab Prometheus. Fill in your details and Helm builds everything.

PostgreSQL

Helm Chart

Redis

Helm Chart

Nginx

Helm Chart

Prometheus

Helm Chart

MongoDB

Helm Chart

Operators = Specialist Departments

An Operator is a SPECIALIST DEPARTMENT. The Percona Operator handles PostgreSQL replication, failovers, backups, and scaling automatically.

Percona Operator

PostgreSQL/MongoDB

Prometheus Op.

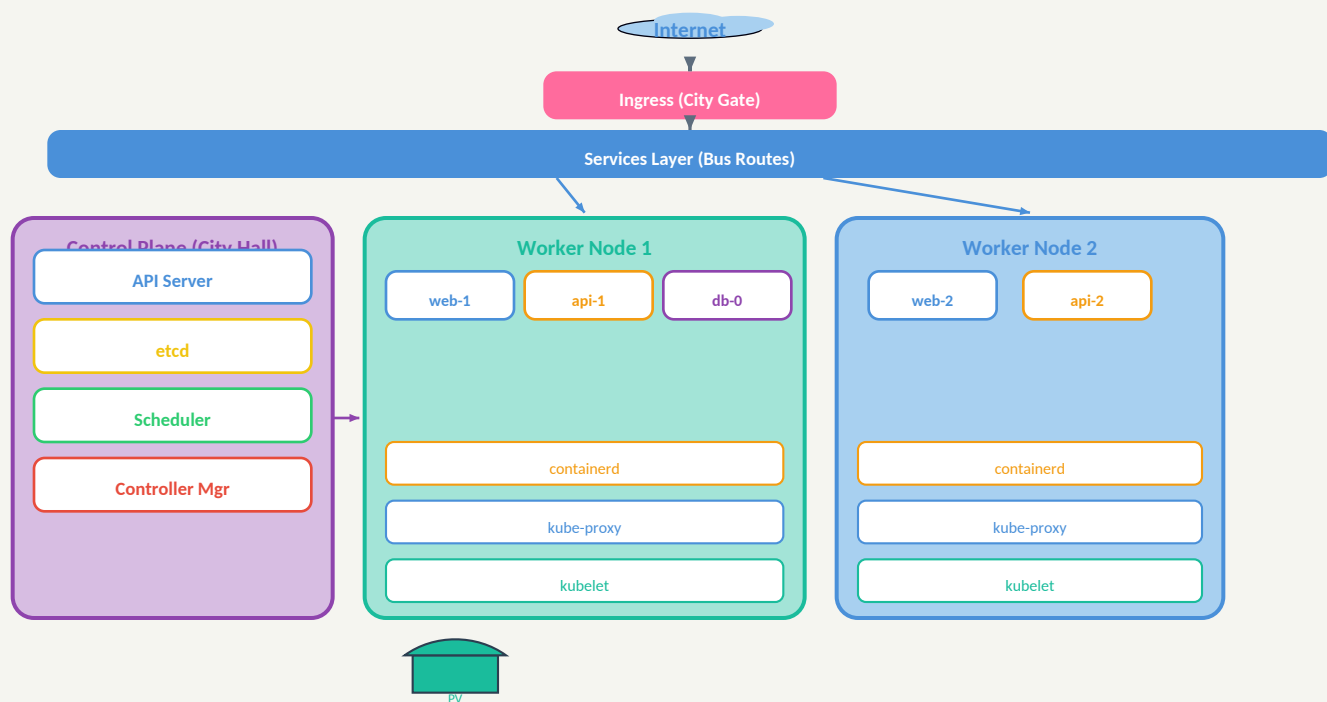
Monitoring Dept

Cert-Manager

Security Certs

16. The Full City Map

KubeCity — Complete Cluster Architecture



ConfigMaps

Secrets

RBAC

NetworkPolicies

Every component working together as a self-managing city